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Off-cycle species are typically considered detrimental to catalytic activity; however, a dormant form can also extend catalyst lifetime. This study shows under which conditions a reversible off-cycle reservoir can increase the turnover number (TON) of a homogeneous catalyst. When both turnover and all deactivation steps are first order in catalyst, branching remains constant over time and the reservoir only holds the catalyst in a dormant state, the distribution of turnovers per catalyst does not change, regardless of the number of reservoirs. The reaction is slowed, and under an external clock, such as a poison that forms independently of the catalyst, the TON decreases. When deactivation is of higher order in catalyst than turnover, a fast reservoir of depth acts as a -fold dilution. This increases the TON up to a ceiling set by the first-order deactivation pathways, and for a given batch time there is an optimal depth. In the published model of (DTBM-SEGPHOS)CuH, the HBpin adduct protects the catalyst from aggregation but is 2.9–5.9 times deeper than optimal. In Hoveyda–Grubbs metathesis, a styrenyl-ether reservoir slows the reaction threefold without reducing the TON. A reservoir can be beneficial only if the catalyst-limited TON decreases as catalyst loading increases, that is, if the slope of log TON against log loading is negative. Dilution experiments and a series with graded amounts of additive then show whether it will be.

Keywords

catalyst deactivation, turnover number, off-cycle species, kinetic modeling, catalyst loading, bimolecular decomposition

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When Can an Off-Cycle Reservoir Raise a Catalyst's Turnover Number?

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Abstract

Off-cycle species are typically considered detrimental to catalytic activity; however, a dormant form can also extend catalyst lifetime. This study shows under which conditions a reversible off-cycle reservoir can increase the turnover number (TON) of a homogeneous catalyst. When both turnover and all deactivation steps are first order in catalyst, branching remains constant over time and the reservoir only holds the catalyst in a dormant state, the distribution of turnovers per catalyst does not change, regardless of the number of reservoirs. The reaction is slowed, and under an external clock, such as a poison that forms independently of the catalyst, the TON decreases. When deactivation is of higher order in catalyst than turnover, a fast reservoir of depth D acts as a D-fold dilution. This increases the TON up to a ceiling set by the first-order deactivation pathways, and for a given batch time there is an optimal depth. In the published model of (DTBM-SEGPHOS)CuH, the HBpin adduct protects the catalyst from aggregation but is 2.9–5.9 times deeper than optimal. In Hoveyda–Grubbs metathesis, a styrenyl-ether reservoir slows the reaction threefold without reducing the TON. A reservoir can be beneficial only if the catalyst-limited TON

decreases as catalyst loading increases, that is, if the slope of log TON against log loading is negative. Dilution experiments and a series with graded amounts of additive then show whether it will be.

Keywords: catalyst deactivation, turnover number, off-cycle species, kinetic modeling, catalyst loading, bimolecular decomposition

Introduction

The number of product molecules generated by each metal center of a homogeneous catalyst before deactivation determines how much metal a process requires and, in many cases, whether the catalyst is viable. This quantity, the turnover number (TON), is governed by deactivation.[1–3] Contemporary microkinetic models incorporate deactivation steps[4,5] or predict catalyst lifetimes.[6] In this literature, off-cycle species are assigned a single function: by occupying a resting state below the catalytic cycle, they increase the energetic span and reduce the turnover frequency (TOF). Consequently, catalyst design generally seeks to minimize such states.[7,8]

Practical experience, however, suggests that this is not always the case, and it is frequently proposed that a dormant form can extend catalyst lifetime. Crabtree[1] describes a deactivation product that can be reactivated as a dormant state. In Hoveyda–Grubbs metathesis, the released styrenyl ether can recapture the active species via a “boomerang” mechanism that “helps to conserve the catalyst charge”[9] and is thought to limit bimolecular coupling of the methylidene.[10] Ligand-free Heck reactions work best at very

low (“homeopathic”) Pd loadings, with most of the Pd present as soluble colloids.[11] In a palladacycle-catalyzed Heck reaction, “Pd lost via deactivation processes may be replaced by Pd metered into the cycle” from a dimer.[12] Ruthenium metathesis catalysts that decompose through bimolecular coupling of methylidenes[10,13] give higher TONs at lower loadings[14] but lower TONs when used as binuclear precatalysts.[15] More recently, Ryan et al.[16] reported that the HBpin adduct of (DTBM-SEGPHOS)CuH, an off-cycle species, “paradoxically, also improves catalyst performance by delaying aggregation”. Related principles underlie dormant chains in living radical polymerization,[17] slow-release boronates,[18,19] enzyme lifetimes[20,21] and protective compartments.[22]

These observations are qualitative and system-specific, and current theory accounts for them only in part. For first-order processes, the number of turnovers before inactivation is a branching ratio, analogous to the partition ratio of a mechanism-based enzyme inactivator[23,24] or to TON viewed as “a selectivity issue”.[2] Gromotka et al. derived a kinetic equation for a linear cycle with both reversible and irreversible deactivation[25] and calculated the lifetime of a catalyst that deactivates irreversibly. They also noted that a high turnover rate protects Rh hydroformylation catalysts by keeping the active complex too scarce to form clusters.[26] Kinetic theories of TON exist for photoinduced catalysis[27] and for electrocatalyst lifetimes,[6] and studies of molecular electrocatalysts define TON but benchmark the TOF.[28–30] None of these addresses under what conditions placing the catalyst in an off-cycle

species increases the TON, by how much, or when simply using less catalyst has the same effect. The distinction matters in practice: a rise in TON at low catalyst loading in photocatalytic CO_2 reduction was attributed to a catalyst reservoir[31] but was later traced to photosensitizer degradation.[27]

In this work, we present this theory and compare it with published kinetic data (Figure 1). The principal findings are as follows. (i) When both turnover and all deactivation steps are first order in catalyst, branching remains constant over time and the reservoir only holds the catalyst, the distribution of turnovers per catalyst is independent of the number of reservoirs. (ii) A fast reservoir of depth D is equivalent to a D-fold dilution. It therefore increases the TON when deactivation is of higher order in catalyst than turnover, up to a ceiling set by the first-order pathways, and there is an optimal depth D. (iii) For a reservoir to be beneficial, the catalyst-limited TON must decrease as catalyst loading increases; this criterion is necessary but not sufficient, and a dilution experiment and a graded-additive series complete the test. Published systems are consistent with the theory.

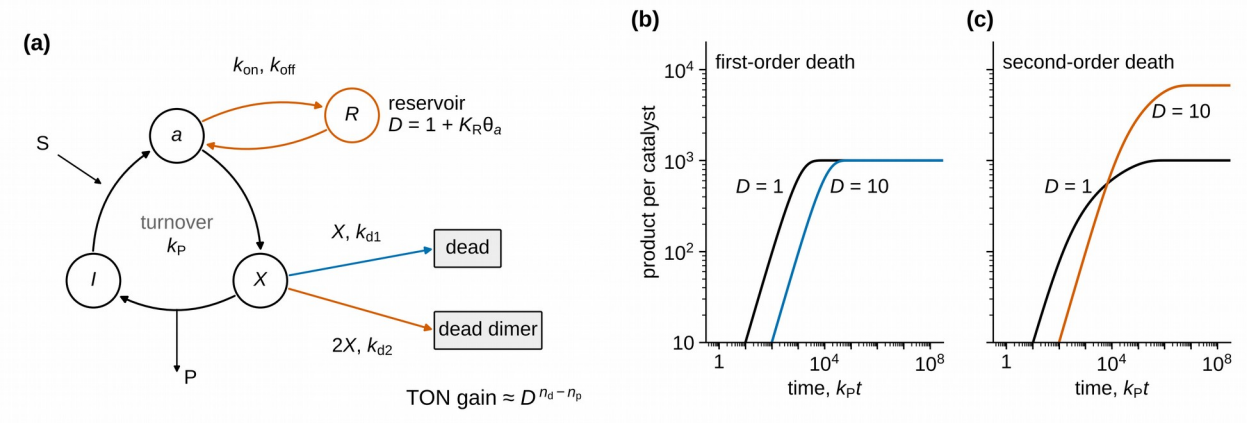


Figure 1. (a) A catalytic cycle with a reversible off-cycle reservoir R on state a (depth $D=1+K_R\theta_a$) and deactivation from X that is either first order (k_{d1}) or second order (k_{d2}) in catalyst. (b, c) Product per catalyst against time without (black) and with a reservoir of depth $D=10$, for (b) first-order and (c) mixed first- and second-order deactivation ($\rho=10^3$), both with $TON_\infty=1000$ without the reservoir. The reservoir slows the reaction in both cases but increases TON_∞ only in (c), by a factor of 6.7.

Methods

Theory and kinetic engine. We followed each catalyst molecule as it moves at random between three kinds of state: on-cycle states, reservoir states, and dead states that it cannot leave. In technical terms, this is a continuous-time Markov chain. We proved the no-go theorem in two independent ways (Section S2). The first removes the reservoir visits from each trajectory, a technique called censoring. The second uses generating functions. We derived the results for the bulk reaction mixture from the standard mass-action rate equations (Section S3). We checked the symbolic identities with SymPy.[32] Every number in the paper was computed with one Python package, tonkit. It takes a reaction network, given either as rate constants or as free energies converted with the Eyring equation, writes out the rate equations and solves them with the LSODA solver[33] in SciPy[34]

(relative tolerance 10^{-8}). We defined second-order deactivation constants by the rate at which catalyst is lost. Some sources define them instead by the rate of the deactivation event, which removes two catalyst molecules, so we converted their values with $k_{d2}=2k_{ev}$. We ran stochastic simulations[35] in two modes. In the first, molecules behave independently, and thinning[36] handles rates that change over time. In the second, molecules interact, which bimolecular deactivation requires; this mode is sped up with Numba.[37] Unit tests in tonkit check every closed-form result against the full kinetic model.

Results and Discussion

Theory

Model and definitions

The path of a single catalyst molecule is traced through a network of on-cycle intermediates, off-cycle species, and irreversible dead states D (Figure 1a). An off-cycle species R is a *pendant reservoir* on the on-cycle state a if it can be entered only from a, returns only to a, and neither forms product nor dies. Its kinetics are otherwise unrestricted and may depend on time, on reagents that are consumed, or on other catalyst molecules (a dimer $2a \rightleftharpoons a_2$ is pendant). With $K_R=[R]/[a]$ and θ_a, the fraction of the on-cycle pool in a, a reservoir in fast exchange, has depth

$D=1+K_R\theta_a$,

the ratio of live catalyst to on-cycle catalyst; $D \approx 1 + K_R$ for a reservoir on the resting state. Experimentally, D is the factor by which a fast reservoir slows product formation. Deactivation proceeds from a state X via first-order (k_{d1}) and second-order (k_{d2}, defined on catalyst consumption, $d[X]/dt = -k_{d2}[X]^2$) pathways, and n_p and n_d denote the catalyst orders of the productive and deactivation fluxes, respectively. We distinguish three turnover numbers: the catalyst-limited TON_∞, set by catalyst death alone; $TON(T)$ at a fixed time horizon T; and the conversion-capped $TON_{cap} \leq S_0/c_0$, which is the value usually reported (Section S1). “Dilution” means a lower catalyst loading at the same substrate concentration.

The no-go theorem

A network is described as *linear* when every turnover and deactivation step is first order in catalyst, with other species entering as pseudo-first-order factors; it satisfies *Condition B* when the probabilities of leaving each state by its different exits do not change with time.

Theorem (no-go). In a linear network that satisfies Condition B, adding any number of pendant reservoirs leaves unchanged the joint distribution of the number of turnovers N that one catalyst completes, the dead end it reaches and the number of times it takes each step; only the time spent in each state changes.

The proof (Section S2) rests on one observation: a visit to a pendant reservoir is an excursion that returns the catalyst to the state it left, with no

product formed and no possibility of deactivation. When these excursions are removed from a trajectory, the remaining sequence of on-cycle and dead states has exactly the transition probabilities of the reservoir-free network, so every function of this sequence has the same distribution; only the timing changes. The mean lifetime increases by $\sum_a K_a \langle T_a^0 \rangle$, where $\langle T_a^0 \rangle$ is the mean time spent in a without reservoirs, and with fast exchange the product curve is stretched in time, $P(t) = P_0(t/D)$. In a linear network, a reservoir therefore adds time but never turnovers. For a single cycle, this result is familiar from the partition ratio of enzyme inactivators,[23,24] from Curtin–Hammett control of product ratios[38] and from the limiting TON of photoinduced catalysis[27] (Section S2.6). The theorem extends it to the whole distribution of N and to any number of reservoirs whose kinetics may depend on time, on consumed reagents or on other catalyst molecules. Stochastic simulations confirm it molecule by molecule (Figure S1).

If any of the hypotheses fails, a reservoir can change the TON. This can happen through a step involving two catalyst units (the order rule, below); an off-cycle species that decomposes, releases into another on-cycle state or reacts with other catalyst species; an additive that alters the branching at a deactivation precursor; or branching that drifts on a clock that the reservoir does not slow (Sections S2.3 and S3.3). Branching that drifts with conversion in a closed batch leaves the mean final yield unchanged, up to corrections of order c_0/S_0 and $1/TON$ (Proposition S3). An external clock,

however, breaks this result. Examples are a reagent that decomposes without catalyst, a poison formed at a fixed rate, or a fixed time horizon. A fast reservoir of depth D makes every external clock run D times faster relative to the reaction, and for a poison that accumulates linearly in time the TON falls as $D^{-1/2}$ (eq S2). Under an external clock, a reservoir therefore lowers the TON.

The order rule: a reservoir is a dilution

Assume that (A1) exchange with the reservoir is fast and its depth is independent of the catalyst concentration (as for a reservoir formed with an additive in excess); (A2) the on-cycle pool is quasi-stationary; (A3) every catalyst flux depends only on the on-cycle pool $F = c/D$; and (A4) the reservoir neither decomposes nor reacts with other catalyst species. A catalyst with a reservoir then behaves exactly like a diluted catalyst without one (Theorem S7):

$$P\left(t; c_0, D\right) = D\, P\left(t/D; c_0/D, 1\right), \quad TON_\infty\left(c_0, D\right) = TON_\infty\left(c_0/D, 1\right).$$

A reservoir of depth D is thus a D-fold dilution that runs D times more slowly and makes D times more product. With $\Phi_d = \tilde{k}_{d1} F + \tilde{k}_{d2} F^2$ (where $\tilde{k}_{d1} = k_{d1} f_X$, $\tilde{k}_{d2} = k_{d2} f_X^2$ and f_X is the fraction of the pool in X) and with $\rho = \tilde{k}_{d2} c_0 / \tilde{k}_{d1}$ the second-order share of deactivation at full loading,

$$G(D)=\frac{TON_\infty(c_0,D)}{TON_\infty(c_0,1)}=\frac{D\ln(1+\rho/D)}{\ln(1+\rho)}\overset{D\to\infty}{\to}\frac{\rho}{\ln(1+\rho)}.$$

The gain grows roughly in proportion to D as long as $D \ll \rho$, and levels off as TON approaches the first-order ceiling $k_p/\tilde{k}_{d1}$. At $\rho=10^3$, a depth of 10 gives a 6.7-fold gain and a depth of 100 a 35-fold gain (not 100-fold), and no depth gives more than 145-fold (Figure 2a). The gain follows a pure power law, $G=D^{n_d-n_p}$, only when $n_d-n_p<1$. With bimetallic turnover and unimolecular death, a reservoir lowers TON as $1/D$.

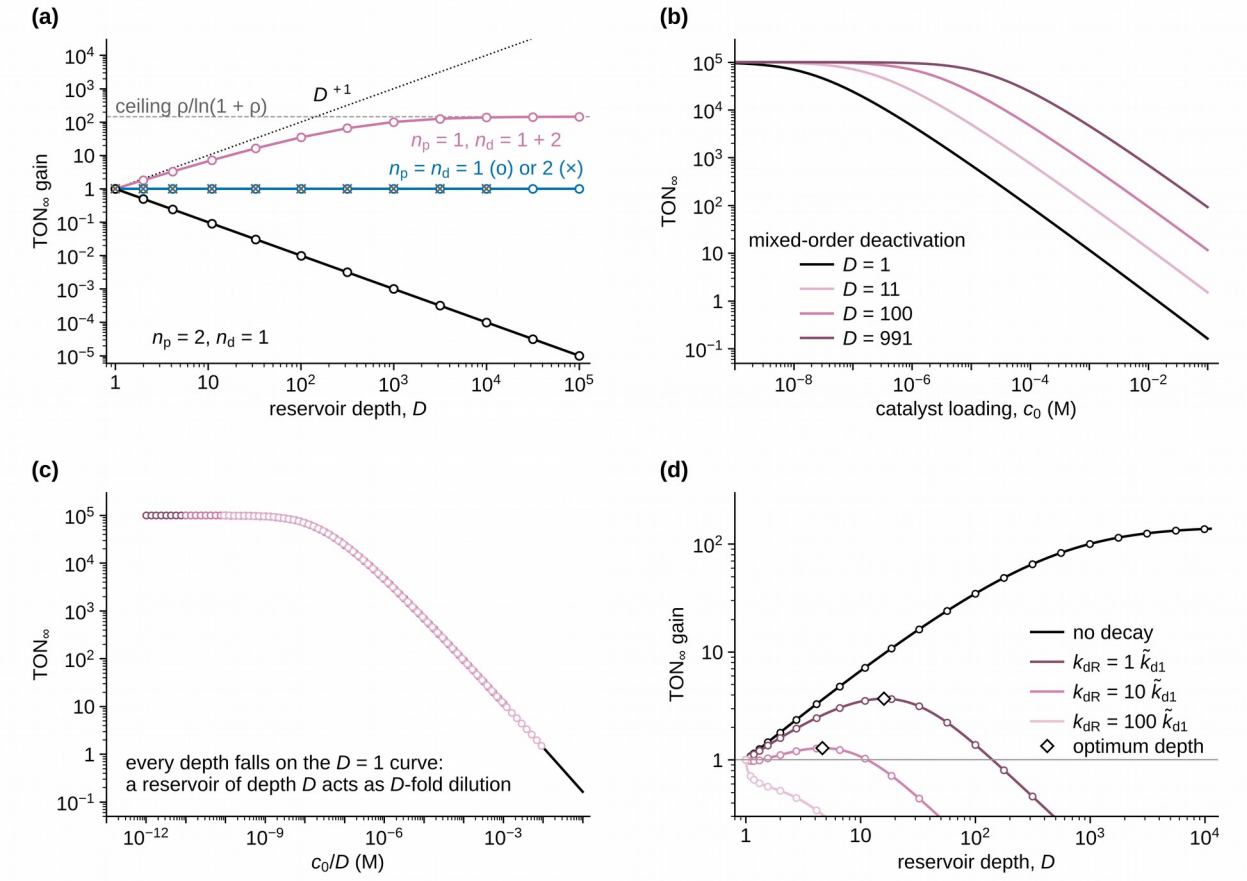


Figure 2. The order rule. (a) Gain in TON_∞ against reservoir depth for different catalyst orders of turnover (n_p) and deactivation (n_d; "1+2", mixed first and second

order with $\rho=10^3$): points, full kinetic model; lines, closed forms. (b, c) TON_∞ against loading for four depths (b) collapses onto one curve when plotted against c_0/D (c). (d) A reservoir that itself decomposes (k_{dR}) has an optimum depth. At $\rho=10^3$, a reservoir that decays ten times faster than the free catalyst dies by its first-order channel gains 1.3-fold at $D^{\prime}=4.6$, after a dip at shallow depth; one that decays 100 times faster gains nothing.

Because a reservoir only rescales the loading axis, plots of TON against c_0 for different depths collapse onto one curve in c_0/D (Figure 2b,c), and

$$s \equiv \frac{\partial \ln TON_\infty}{\partial \ln c_0} = \frac{-\partial \ln TON_\infty}{\partial \ln D} = -\left(n_d - n_p\right)_{eff}.$$

Under A1–A4, the order rule and the loading diagnostic used below are equivalent. The converse can fail, however: if the death hazard per turnover grows with an accumulating by-product, $TON_\infty=\left(2k_P/k_B c_0\right)^{1/2}$ and $s=-1/2$, but a reservoir leaves TON unchanged while dilution raises it $\sqrt{D}$-fold (Section S3.6). A negative slope is therefore necessary for a reservoir to help, but not sufficient.

A reservoir that decomposes ($R \to D$, k_{dR}) has an interior optimum, $D^{\prime} \approx 0.505\sqrt{\tilde{k}_{d2} c_0/k_{dR}}$, because its lifetime acts as a horizon. TON can dip at shallow depth and still rise at greater depth, so whether such a reservoir helps depends on the TON at the optimum rather than on the initial slope (Figure 2d). Reservoirs that react with the active species, and dimeric reservoirs, are treated in Section S3.3.

Optimal depth; reservoir, dilution or slow addition?

A deeper reservoir also slows the reaction more. Over a horizon T with pure second-order death, $TON(T;D)=\left(k_P D/\tilde{k}_{d2}c_0\right)\ln\left(1+\beta/D^2\right)$ with $\beta=\tilde{k}_{d2}c_0 T$, and the maximum lies at

$D^{\dagger}=0.505\sqrt{\tilde{k}_{d2}c_0 T}$;

and a reservoir helps only if $\beta>3.92$ (Section S3.4, Figure S2; the prefactor becomes 0.714 for a constant defined on the event rate). A useful benchmark is the product that a constant pool would make while leaving no catalyst at T, which bounds every strategy with the same metal budget. A fast reservoir at $D^{\dagger}$ reaches 80.5% of this bound. A catalyst charged in its dormant form and released at a first-order rate reaches 87% at $\beta=10$ and 90% for $\beta\gtrsim 10^3$, and constant slow addition approaches the bound. With slow exchange, the catalyst should be charged in its dormant form or pre-equilibrated with the additive, because a catalyst charged fully active suffers an unprotected burst of second-order death at $t=0$.

This also answers the objection that one could simply use less catalyst. Without a horizon, dilution and a reservoir give the same TON_∞ at the same pool concentration (eq 2), but dilution reaches it with less metal and therefore makes less product. At a fixed horizon, dilution raises TON per metal only at the expense of product per batch. At a TON per metal of 1% of

$k_P T$, dilution makes 6.5 units of product, a fast reservoir 65, slow release 81 and constant addition 99, against a bound of 100. When both first-order ($\gamma = \tilde{k}_{d1} T$) and second-order ($\beta$) deaths are possible within T, the best reservoir improves a batch by more than 5% only if $\rho \gtrsim 0.1$ and $\beta \gtrsim 3$ ($\beta \gtrsim 10$ when first-order death is negligible within T).

Additives: parking and selectivity

An additive that binds the catalyst can change TON in two ways (Figure S3). As a pendant reservoir, it acts through the population it parks, $D = 1 + K_R \theta_a$, rather than through its free energy, and only when $n_d > n_p$. In an energetic-span profile, for example, a reservoir on an intermediate that holds 1.5×10^{-9} of the pool has no effect, even though its free energy lies below that of the turnover-determining intermediate. As a ligand L that enters the transition states, the additive changes the branching between turnover and death. In a linear network with one dominant transition state on each side,

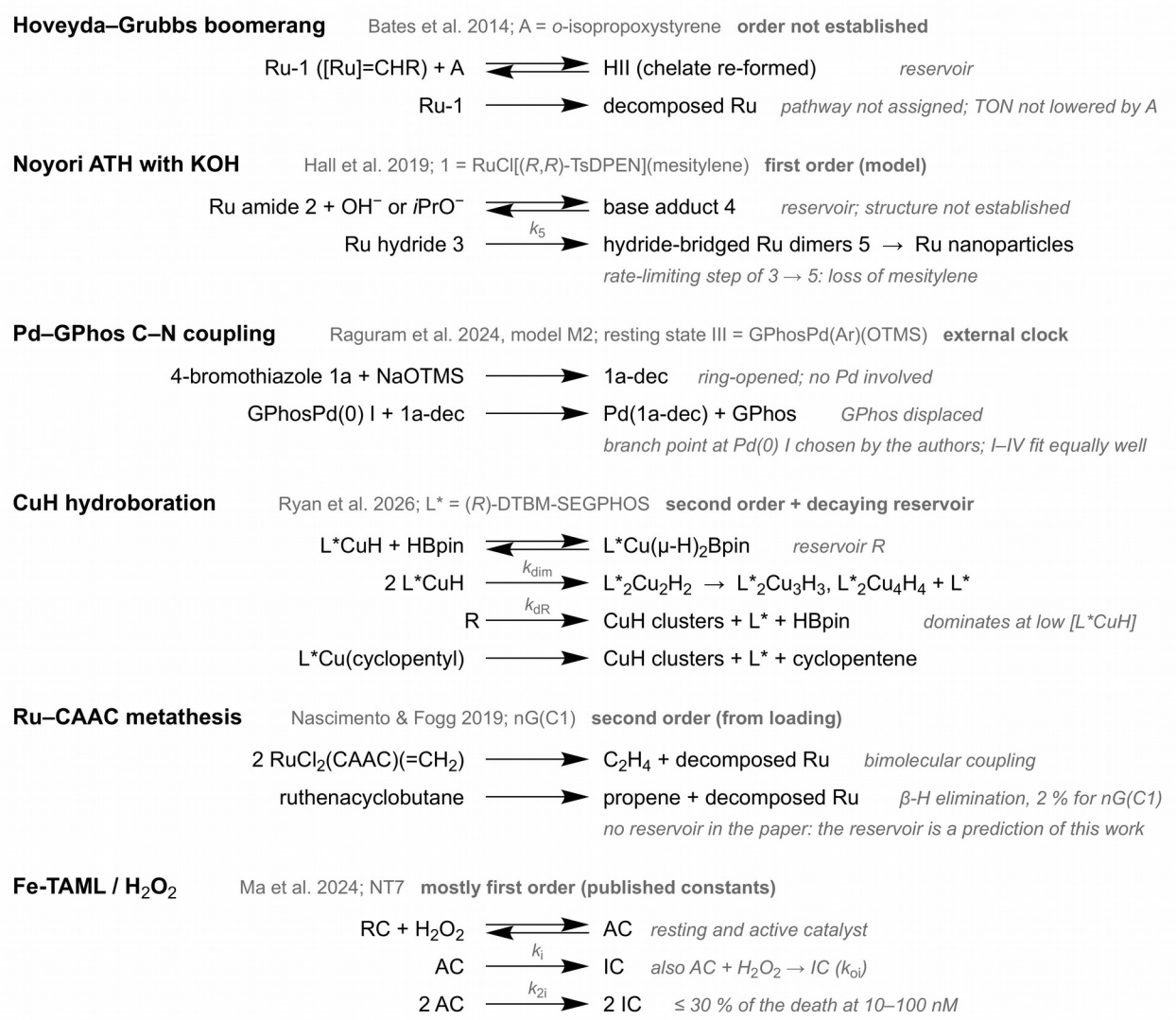
$$\frac{\partial \ln TON_\infty}{\partial \ln [L]} = n_L(TDTS) - n_L(T S_d),$$

that is, the difference in L content between the turnover-determining transition state (TDTS) and the deactivation transition state, independent of the resting state and of any adduct. Deactivation that requires ligand loss, as proposed for Rh/PPh$_3$ hydroformylation and monoligated Ni(0),[39,40] gives +1, and donor-promoted decomposition of Ru methylidenes[41] gives -1. Both

results follow from a degree of TON control, defined by analogy with the degree of rate control,[7,42] which is zero for every intermediate of a linear network (Section S2.5).

Case studies

We compared the theory with published systems in which a reversible off-cycle species and a deactivation channel have been characterized (Scheme 1). We first ran each published model with its own constants against its own data. Fits are graded as quantitative ($R^2 \geq 0.999$ for substrates and products, ≥ 0.95 for catalyst species, residuals within a few times the point-to-point noise), semi-quantitative or qualitative, and every constant is traced to its source (Sections S5.1–S5.6; summary in Figure S4).



Scheme 1. Reversible off-cycle reservoirs and deactivation routes in the systems studied, drawn as labeled formulas from the original papers. Tags give the catalyst order of deactivation relative to turnover where it has been established; gray notes give the status of each step; the Ru-CAAC reservoir is a prediction of this work. L* = (*R*)-DTBM-SEGPHOS; HII = second-generation Hoveyda–Grubbs catalyst.

First-order deactivation and external clocks

Hoveyda–Grubbs metathesis with a styrenyl-ether reservoir (measured). Bates et al.[9] added 1 equiv of the styrenyl ether released on initiation of HII to the cross-metathesis of anethole with methyl acrylate. The ether recaptures the alkylidene and re-forms the chelated precatalyst.

In their Fig. 1b (our Figure 3a), the control stops at 77% conversion (TON 1540, cap 2000). With the ether added, the same conversions are reached about 3.1 times later and, after 4 h, the TON is 0.96 of that of the control and still rising (Figure 3b); the authors noted that TON “was unaffected by added A”. The reservoir therefore slows the reaction threefold without costing any turnovers. It cannot, however, show that decomposition is first order. Because the control stops at 77% of its cap, even a substantial second-order share ($\rho = 0.5$–2) would raise the observed TON by only 6–13%. Furthermore, recapture can intercept the methylidene that couples,[10] so 3.1 need not be the depth seen by the decomposing species (Section S5.3). A decisive experiment would run the control far from its cap and measure the TON and the slowdown factor for several amounts of ether.

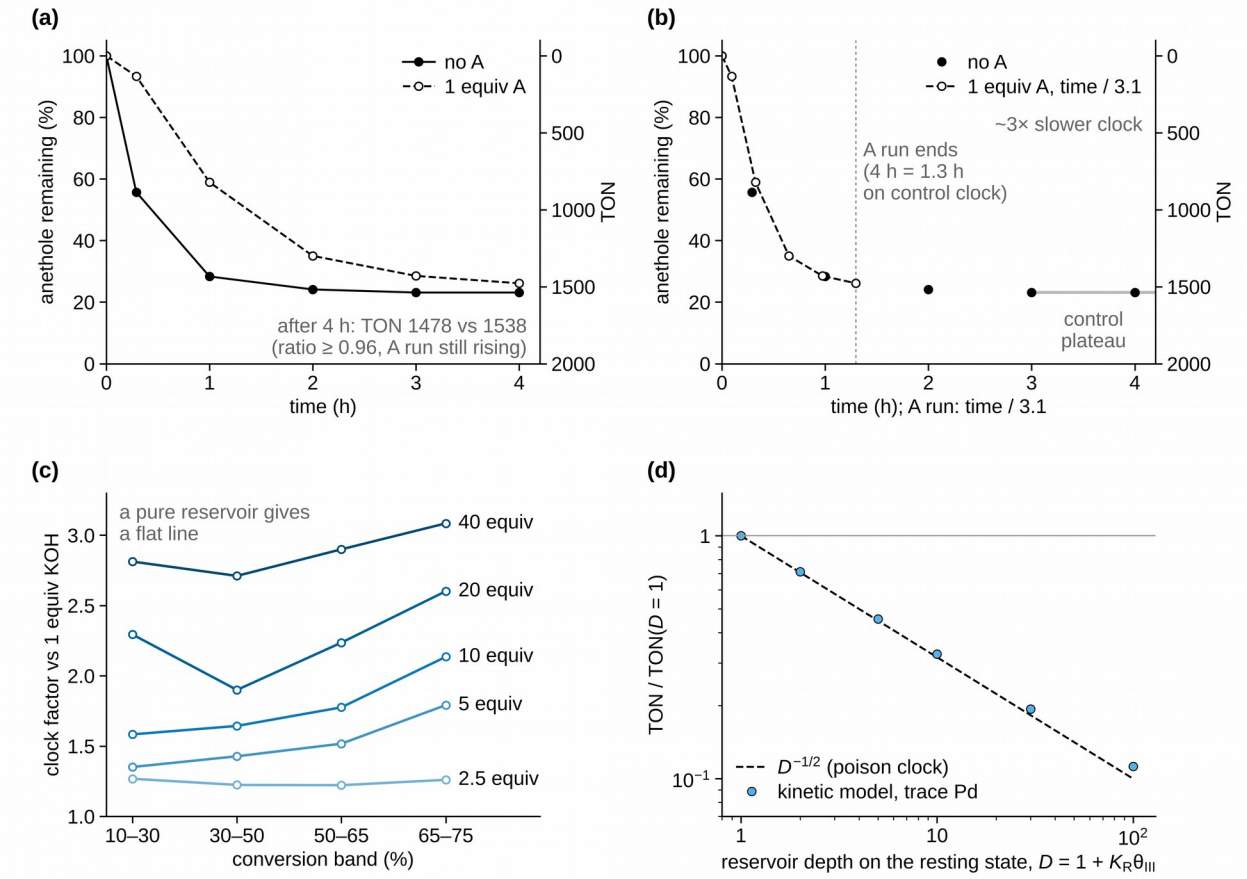


Figure 3. First-order deactivation and an external clock. (a) Cross-metathesis of anethole with methyl acrylate (0.05 mol% HII, 6 equiv acrylate, 1,2-dichloroethane, above 50 °C, not specified further) without (filled) and with (open) 1 equiv of styrenyl ether A; points digitized from ref 9. (b) The same points with the time axis of the A run divided by 3.1. The A run ends (dashed line) after 1.3 control-equivalent hours, before the control reaches its plateau (gray, from 3 h), so its TON ratio of 0.96 is a lower bound. (c) Noyori ATH with 2.5–40 equiv KOH: factor by which each run is slower than the 1-equiv run at matched conversion (FlowNMR data of ref 43); a pure reservoir would give a constant factor. (d) Under an external clock (a poison formed from the substrate by the base; network M2 of ref 44), a reservoir of depth D on the resting state lowers the TON at trace Pd to close to $D^{-1/2}$ of its value without the reservoir.

Published models: transfer hydrogenation and an external clock. The kinetic model of a Noyori TsDPEN–Ru transfer hydrogenation inhibited by excess KOH through a base adduct[43] obeys the theorem exactly: for 1–40 equiv KOH the distribution of N is unchanged, and only t_{50} increases. The FlowNMR runs cannot test this, because they end near equilibrium long

before the catalyst dies. Compared at matched conversion, they mostly show a change of time scale (by a factor of 1.2 to 3.1), together with a 10–35% drift that a reservoir of constant depth cannot produce (Figure 3c; Section S5.2). In the amination of 4-bromothiazole catalyzed by Pd with the dialkylbiaryl phosphine GPhos (Pd-GPhos), the base NaOTMS converts the substrate into a species that displaces GPhos from Pd(0).[44] This poison is formed without Pd and thus acts as an external clock. In the authors' network (model M2), a hypothetical fast reservoir on the resting state lowers the TON at trace Pd to 0.33 at $D=10$, close to $D^{-1/2}=0.32$ (Figure 3d). This result rests on the structure of the network; the printed constants run 1.5–2 times faster than the published traces (Section S5.4). The same paper also provides the experimental counterpart: slow addition of the bromothiazole "resulted in a marked improvement in the yield", slowing the poison clock relative to the catalyst, the reverse of what a reservoir does. A species that shields the vulnerable Pd state from the poison, as the authors propose for the ethereal solvent, is not a pendant reservoir and can raise the TON.

A reservoir against aggregation: (DTBM-SEGPHOS)CuH (published model)

Ryan et al.[16] directly observed the (DTBM-SEGPHOS)CuH monomer and, with full time courses, measured its insertion into alkenes, its dimerization and aggregation, and its binding of HBpin to an off-cycle adduct LCu(μ-H)$\square_2$

Bpin that slowly loses ligand. Free LCuH deactivates by second-order aggregation, the adduct and the alkyl by first-order ligand loss, and turnover is first order. The order rule therefore applies, with a reservoir that decays and drains as HBpin is consumed. Rerun on the time courses in their Supporting Information, the published constants give $R^2 \geq 0.9997$ for substrates and products and ≥ 0.979 for the copper species (Figure 4a; Section S5.1). No single set of first-order constants describes all five datasets, so everything below is a prediction of the authors' model, carried through seven constant sets.

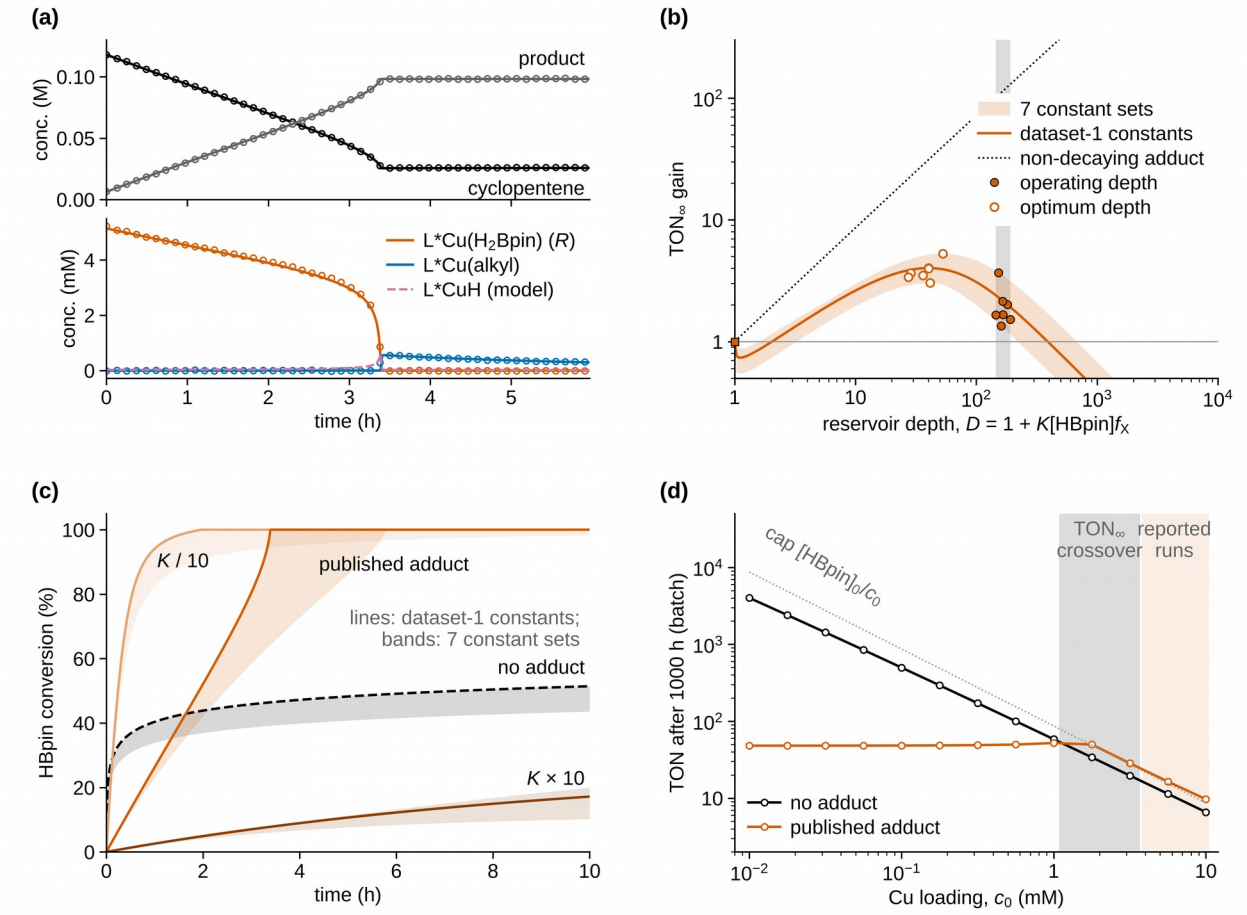


Figure 4. The HBpin adduct of (DTBM-SEGPHOS)CuH as a reservoir against aggregation (data and model of ref 16). (a) Published model (lines) compared with

the raw time course (points), dataset 1. (b) Gain in TON_∞ (substrates held at their dataset-1 concentrations) against depth for seven constant sets (band; line, dataset-1 constants). The initial operating depth (filled, D=145–192) is 2.9–5.9 times greater than the optimum (open, $D^{¿}$=27–53) because the adduct decomposes; dotted, a non-decaying reservoir; square, D=1. (c) Simulated batch (dataset-1 conditions): without the adduct, conversion stops at 44–51%; a tenfold weaker adduct reaches full conversion 2.2–4.5 times faster. (d) TON after 1000 h in the dataset-1 batch against copper loading: the curves cross at 1.1 mM; band, crossover of TON_∞ over the seven constant sets (1.1–3.7 mM); the experiments used 3.8–19 mM.

Within this model, the HBpin adduct is a genuine reservoir against aggregation. Without it, the simulated batch stops at 44–51% HBpin conversion because the copper aggregates; with it, the reaction proceeds to completion (Figure 4c). The loading slope of TON_∞ ranges from -0.92 to -0.95 without the adduct and from -0.05 to -0.17 at the operating depth: the reservoir has moved the catalyst almost into the no-go regime. However, the adduct is not optimal. Its initial depth, D=145–192, is 2.9–5.9 times greater than the optimum $D^{¿}$=27–53 set by its own decay, so it raises TON_∞ 1.3- to 3.7-fold, whereas 3.0- to 5.3-fold was available (Figure 4b). The optimum is uncertain by about a factor of two, but the adduct is too deep for every constant set. Because the reported runs are conversion-limited, only simulated batches can be compared with experiment. A tenfold weaker adduct reaches full conversion 2.2–4.5 times faster, whereas a tenfold stronger one stalls at 10–20% conversion; a series with varied HBpin excess or a more weakly binding ligand would test this. The adduct pays off only above a crossover loading (1.1–3.7 mM Cu for TON_∞, 1.1 mM in the dataset-1 batch; Figure 4d), which lies below the 3.8–19 mM used in the experiments.

Reading the deactivation order from loading data

Equation 4 provides a test that does not require knowledge of rate constants: the catalyst-limited TON is measured at two or more loadings. A slope of $s \approx 0$ indicates that no reservoir can increase TON, whereas a slope of $s < 0$ suggests that a reservoir may be effective. In this case, the dilution test from equation 2 predicts the gain, assuming conditions A1–A4 are met and no other factors depend on loading (Table 1). All reservoir depths align on a single master curve of s versus $c_0/(D\,c^{\ddagger})$, where $c^{\ddagger} = \tilde{k}_{d1}/\tilde{k}_{d2}$ represents the crossover loading (Figure 5a).

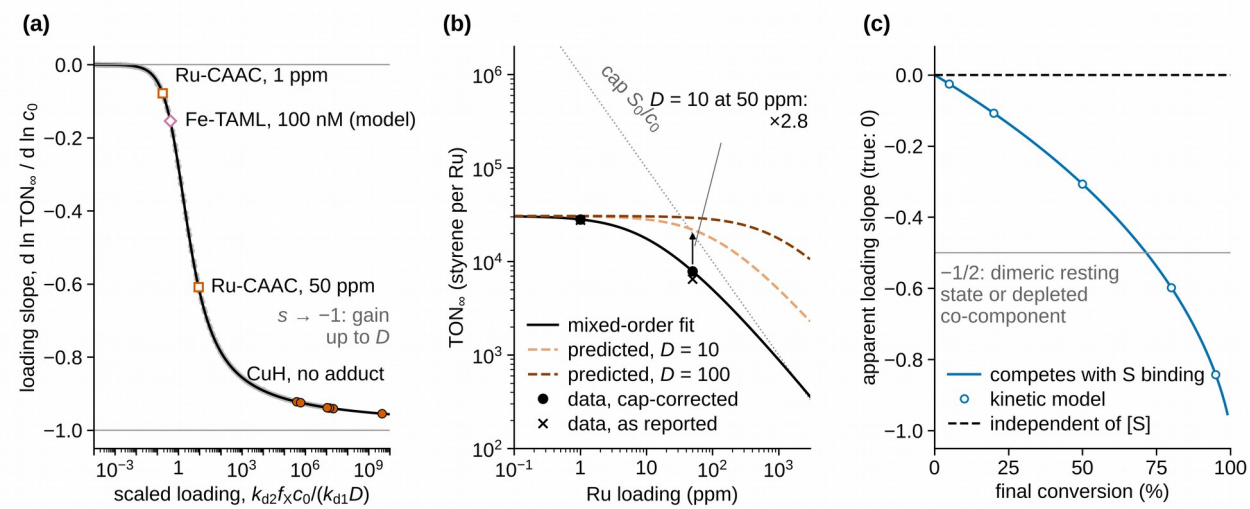


Figure 5. The loading diagnostic. (a) Slope $s = \partial \ln TON_\infty / \partial \ln c_0$ against scaled loading. All reservoir depths fall on a single curve, and each case sits where its slope places it (Fe-TAML at 100 nM from the published constants). A reservoir can raise TON only where $s < 0$. (b) Ru-CAAC metathesis (data of ref 14): the two loadings fix the mixed-order model (black), which predicts the gain from a reservoir of depth 10 or 100 (colored). (c) Negative slopes that do not come from second-order deactivation: conversion capping when deactivation competes with substrate binding, and a depleted co-component or a dimeric resting state ($-1/2$).

Table 1. Effects other than the catalyst order of deactivation that make TON depend on loading, and how to detect them.

Effect	**Effect on s**	**Check**
conversion capping	negative even for first-order death (-0.11 at 20%, -0.31 at 50% conversion when death competes with substrate binding)	final conversion below about 20%, or correct from the time course up to about 50%
fixed horizon	understates $\lvert s \rvert$ unless $\beta \gg 1$	follow runs to the death plateau
product inhibition, equilibrium	false plateau	spike with fresh substrate and, separately, fresh catalyst
death driven by a by-product or volatile co-product	$s<0$ with no reservoir gain	spike the by-product at fixed loading; compare open and closed vessels
poisons and impurities in substrate or solvent	$s>0$ until the poison is titrated	vary purification at fixed loading; add a scavenger
loading-dependent activation of a precatalyst	s reflects the active fraction	measure the active catalyst, or use a preactivated form
a co-component runs out (photosensitizer, oxidant)	$s=-1/2$ without catalyst death	check what is left at the end
dimeric resting state	s drifts toward $-1/2$	speciation against loading
reservoir is not a dilution	dilution test over-predicts the gain: a reservoir leaves activation, impurities and co-reagents at c_0	graded additive series with measured clock factor

Ru-CAAC metathesis (prediction). For styrene self-metathesis with the CAAC catalyst nG(C1) (0.20 M styrene; ppm = mol Ru per mol styrene), Nascimento and Fogg[14] report TONs of 27 800 at 1 ppm and 6500 at 50 ppm (10 μM Ru). The methylidene of such catalysts decays by bimolecular coupling.[13] Once the 50-ppm value is corrected for conversion, the slope is -0.33, and the two loadings fix the mixed-order model (eq 3), with a first-order ceiling $k_P/\tilde{k}_{d1}=30\,700$ and $c^{\ddagger}=1.1$ μM (5.6 ppm). A reservoir of depth 10 on the coupling species would then raise TON 2.8-fold at 50 ppm, and no depth would give more than 3.9-fold (Figure 5b). This prediction rests on

two data points and could be wrong in either direction. The 50-ppm run makes about twelve times more ethylene, which feeds the methylidene that couples, and the 24-h TON at 1 ppm may be partly a horizon TON (Section S5.5). Measurements at a third and a fourth loading, with time courses followed to a plateau and ethylene removed, would test the model before any additive is made. One candidate reservoir is an N-donor that binds the methylidene reversibly[10] without promoting its decomposition.[41]

Fe-TAML: a capped slope. For an Fe-TAML catalyst at 10 and 100 nM,[45] the reported TONs reach 62% and 97.5% of their respective substrate caps, so their ratio of 0.16 is mainly due to conversion capping. The data neither confirm nor exclude a loading effect, and the published deactivation constants predict little bimolecular deactivation at these loadings (Section S5.6).

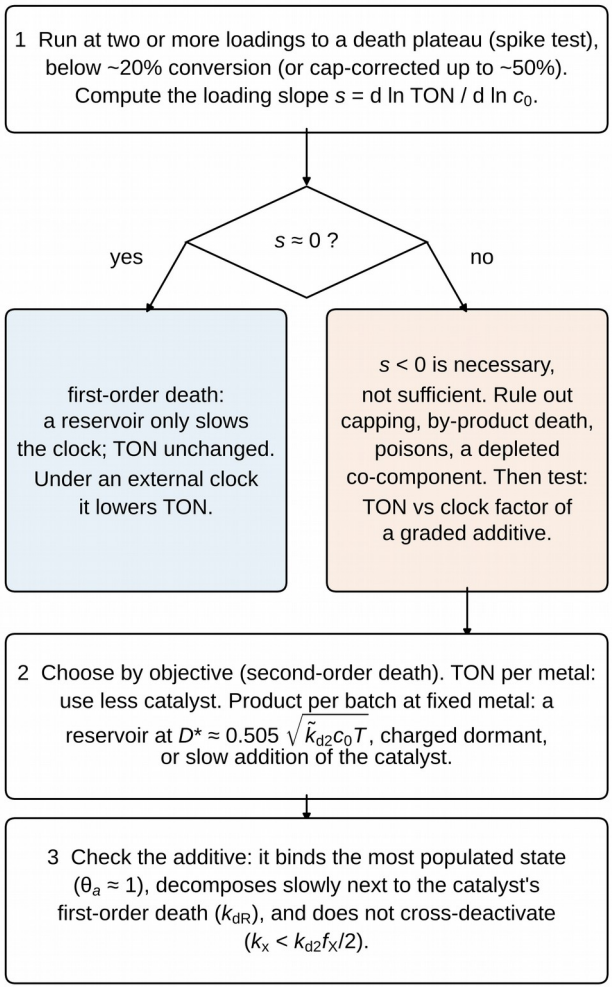
Design rules

How well the cases test the theory. Every case falls where the theory places it, but the cases test it to different degrees (Figure S4). For first-order deactivation, the theorem holds exactly in the published ATH model, and the boomerang experiment shows a threefold slower reaction with no loss of TON, although it does not establish the order of decomposition. For bimolecular deactivation, the published CuH model gives a 1.3- to 3.7-fold gain from the HBpin adduct as it operates, and two loadings predict a 2.8-fold gain for a depth-10 reservoir in Ru-CAAC metathesis. Under an external

clock, the Pd-GPhos network loses TON as $D^{-1/2}$, in line with the authors' slow-addition result. No study has yet varied a reservoir of known depth and measured the effect on TON, and this is the experiment the theory calls for.

What to measure, and what can go wrong (Scheme 2, Table 1). First, measure the catalyst-limited TON in duplicate at two or more loadings spanning at least an order of magnitude, and follow each run to a plateau that a spike test confirms is due to catalyst death. Keep final conversions below about 20%, where capping shifts s by at most about 0.1, or correct for capping from the time course up to about 50% conversion. Above that, a reported TON says little about deactivation. If TONs are known to $\pm 10\%$, two loadings a factor of ten apart fix s to about ± 0.06, and a third loading shows whether the slope is constant. Second, interpret the slope. If $s \approx 0$, a reservoir can only slow the reaction, and under an external clock it lowers TON. If $s<0$, rule out the other causes listed in Table 1, then run the dilution test (c_0/D without additive) and, as the decisive test, a graded additive series. A reservoir helps only if TON rises with the slowdown that each amount of additive produces. Third, choose according to the objective. To maximize TON per metal, dilute. To maximize product per batch for a fixed metal budget (second-order death, no other clock), use a reservoir near $D^{\dagger}$, a dormant precatalyst with a release half-life near $0.28\,T$, or slow addition. The constant $\tilde{k}_{d2} = k_P / \left(\tau_1 c^{\dagger}\right)$ can be obtained from the loading data, with $\tau_1 = k_P / \tilde{k}_{d1}$ the first-order ceiling. Fourth, check the additive. It must bind the

populated state, must not decompose faster than it protects, and must not react with the active species faster than $k_{d2}f_X/2$.



Scheme 2. Procedure for deciding whether an off-cycle reservoir, dilution or slow addition will increase the turnover number.

Variable time normalization analysis determines the order in catalyst[46] and, in its extended form, catalyst activation and deactivation from the reaction profiles;[47] same-excess experiments detect deactivation and product inhibition.[48] Because these methods use complete time courses, they are the more reliable test where such profiles are available. The TON slope needs

only end points, but it translates directly into the gain that a reservoir can give; the two approaches should agree.

When not to minimize off-cycle species. The usual advice to avoid off-cycle species because they reduce the TOF[7,8] is valid in the first-order regime, where a reservoir costs rate and gains nothing. In the second-order regime, however, this advice is incomplete: there the same species trades rate for lifetime, and the balance can be tuned, although in practice the rate lost, the heat released and the selectivity often decide. In the CuH model, the HBpin adduct is what allows the reaction to go to completion, yet a weaker adduct, obtained either by ligand design or by changing the HBpin concentration, would give the same yield in a shorter time. Wherever bimolecular deactivation and reversible off-cycle species are known but TON has not been measured as a function of loading, it is worth asking whether deactivation is of higher order in catalyst than turnover. Examples include additives that bind monoligated Ni(0) and inhibit its irreversible dimerization,[40] dormant Rh dimers in hydroformylation,[39] “homeopathic” Pd loadings,[11] palladacycle dimers in Heck chemistry,[12] and site isolation on a support[49] or in compartments,[22] all of which act as a very deep reservoir or an extreme dilution.

Conclusions

A reversible off-cycle reservoir that only holds the catalyst in a dormant state can increase the turnover number (TON) only if deactivation is of

higher order in catalyst than turnover. If both turnover and all deactivation steps are first order and branching remains constant over time, any number of pendant reservoirs, whatever their kinetics, leave the whole distribution of turnovers per catalyst unchanged. They slow the reaction, and under an external clock they reduce the TON. When deactivation is bimolecular, a fast reservoir of depth D is exactly equivalent to a D-fold dilution; it increases the TON approximately in proportion to D, up to the ceiling set by the first-order pathways, with an optimal depth of $0.505\sqrt{\tilde{k}_{d2}c_0T}$ over a horizon T. Published kinetic studies are consistent with these findings but test them only in part. For a reservoir to be beneficial, the catalyst-limited TON must decrease as catalyst loading increases. A dilution run and a graded-additive series then tell the experimentalist whether it will be, and whether a reservoir, dilution or slow addition is the best choice.

Associated Content

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acscatal.XXXXXXX.

Notation and the three TON definitions; proofs of the no-go theorem, the conditions under which it fails and the degree of TON control, with a method check against ref 27; closed forms (order rule and its variants, optimal depth, slow release, dosing bound, additive composition rule,

loading diagnostic) with the figures on the no-go theorem, the optimal depth and additives (Figures S1, S2 and S3); the numerical engine, stochastic simulations and parameter sweeps; each case-study model with its constants, reproduction and reservoir variation, and a summary of the cases (Figure S4).

Data Availability Statement

The `tonkit` package is available at https://github.com/joshazerty/tonkit (release 0.1.0)

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Notes

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